Setup: task, score, data

Predict the count of each bike type at a station, h minutes ahead

- Task: for every station and time t, predict the number of classic bikes and of e-bikes at t + h, h = 2 to 15 min. One model for all stations; h is an input.
- Inputs (20 numbers + station): h; classic, e-bike and dock counts now, 10 and 30 min ago; renting / returning now; capacity; lat / lon; hour (sin, cos); weekend; temperature; humidity. The station enters through a learned 8-number embedding.
- Loss: mean squared error (MSE) of the count. The net predicts the change; forecast = count now + change.
- Score: skill = 1 − MSE_model / MSE_persistence, on the same rows.
  - Persistence = doing nothing: the count at t carried forward.
  - 0 = no better than doing nothing; 1 = perfect; below 0 = worse than doing nothing.
- Low rows: count at t ≤ 2, the hard cases the team table was built around.

Split	Days (local)	Rows	Used for
train	Sep 29 – Oct 18	7,525,845	fitting
val	Oct 19 – 23 (Mon–Fri)	2,083,617	early stopping, model choice
test	Oct 24 – 28 (from Sat)	1,747,660	one score, after the choice

Data file: team repo main 969473f build, SHA-1 263bb791… (checked). Persistence MSE on val: classic 0.393, e-bike 0.150 bikes².

What was tested

24 configurations, 68 runs; ranked at 4 epochs, finalists trained fully with 3 seeds

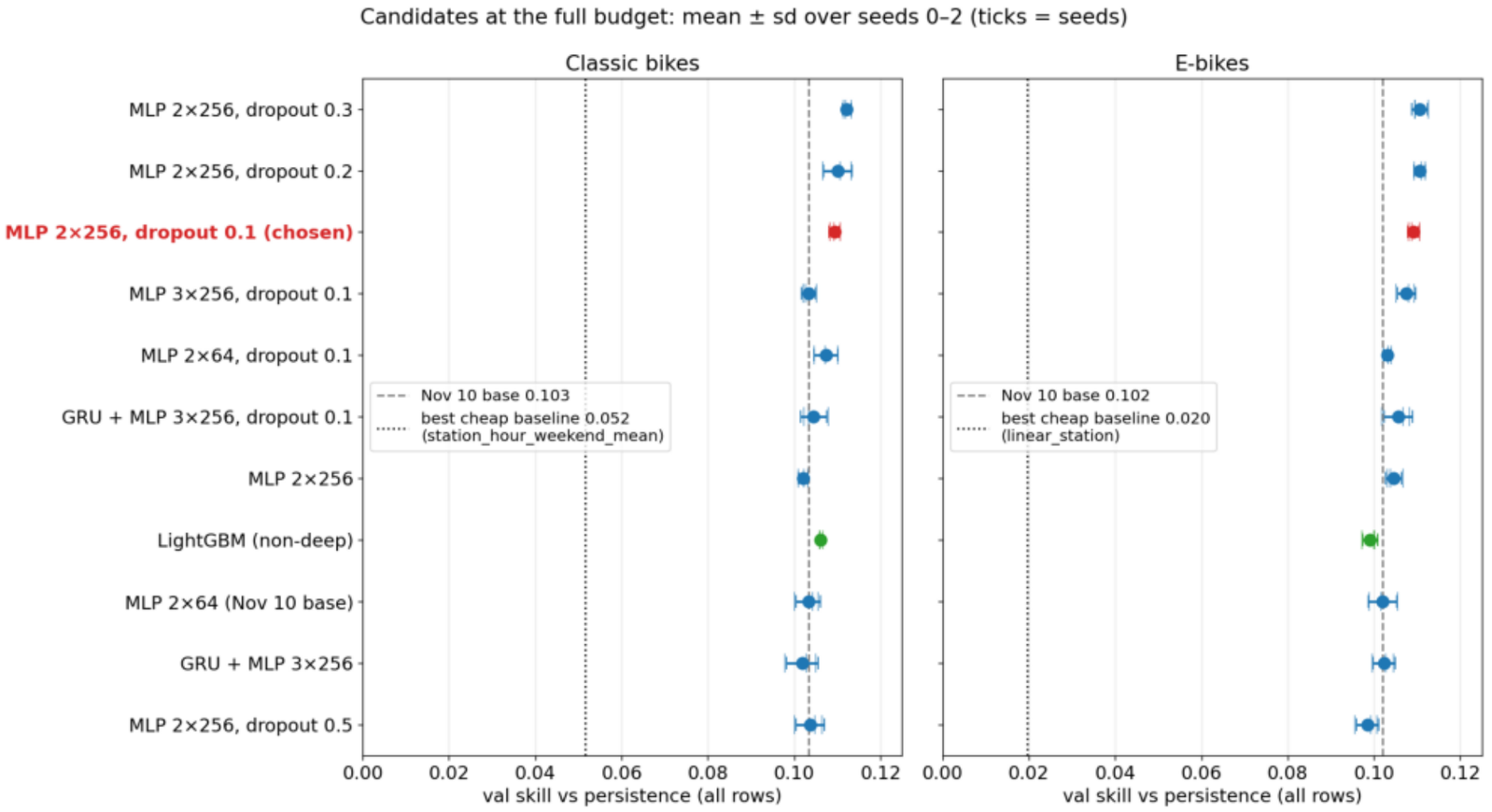
- Architectures:
  - MLP widths and depths: 2×64 (Nov 10 base), 2×128, 2×256, 3×128, 3×256, 4×256
  - station embedding 8 or 32
  - GRU over the lag counts (t − 30, t − 10, t) + MLP head
  - LightGBM (gradient-boosted trees) as a non-deep reference
- Hyperparameters (on 2×256):
  - learning rate 3e-4, 1e-3, 3e-3; batch 1024, 4096, 16,384
  - weight decay 1e-4; dropout 0, 0.1, 0.2, 0.3, 0.5
- Why no LSTM: the table has only 3 lag steps. The GRU (the same gated-RNN family) gave no gain over an MLP of the same size, with or without dropout. A longer history would be needed (untested).

Val skill, 3 seeds	classic	e-bike	classic low	e-bike low
MLP 2×64 (base)	0.103 ± 0.003	0.102 ± 0.003	0.088 ± <0.001	0.075 ± 0.001
MLP 2×64, drop 0.1	0.107 ± 0.003	0.103 ± 0.001	0.090 ± 0.003	0.076 ± 0.001
MLP 2×256	0.102 ± 0.001	0.105 ± 0.002	0.084 ± 0.002	0.076 ± 0.001
MLP 2×256, drop 0.1 ★	**0.109 ± 0.001**	**0.109 ± 0.001**	**0.092 ± 0.002**	**0.080 ± 0.002**
MLP 2×256, drop 0.2	0.110 ± 0.003	0.111 ± 0.001	0.093 ± 0.001	0.081 ± 0.001
MLP 2×256, drop 0.3	0.112 ± 0.001	0.111 ± 0.002	0.093 ± 0.001	0.080 ± 0.001
MLP 2×256, drop 0.5	0.104 ± 0.003	0.098 ± 0.003	0.085 ± 0.004	0.073 ± 0.003
MLP 3×256, drop 0.1	0.103 ± 0.002	0.108 ± 0.002	0.086 ± 0.004	0.077 ± 0.001
GRU + 3×256	0.102 ± 0.004	0.102 ± 0.003	0.085 ± 0.002	0.077 ± 0.003
GRU + 3×256, drop 0.1	0.105 ± 0.003	0.106 ± 0.003	0.087 ± 0.001	0.078 ± 0.002
LightGBM	0.106 ± <0.001	0.099 ± 0.002	0.094 ± 0.002	0.080 ± 0.001
best cheap baseline	0.052	0.020		

★ chosen. Source: model_search_summary.csv (38_search_report.py)

Which wins, and by how much against the noise

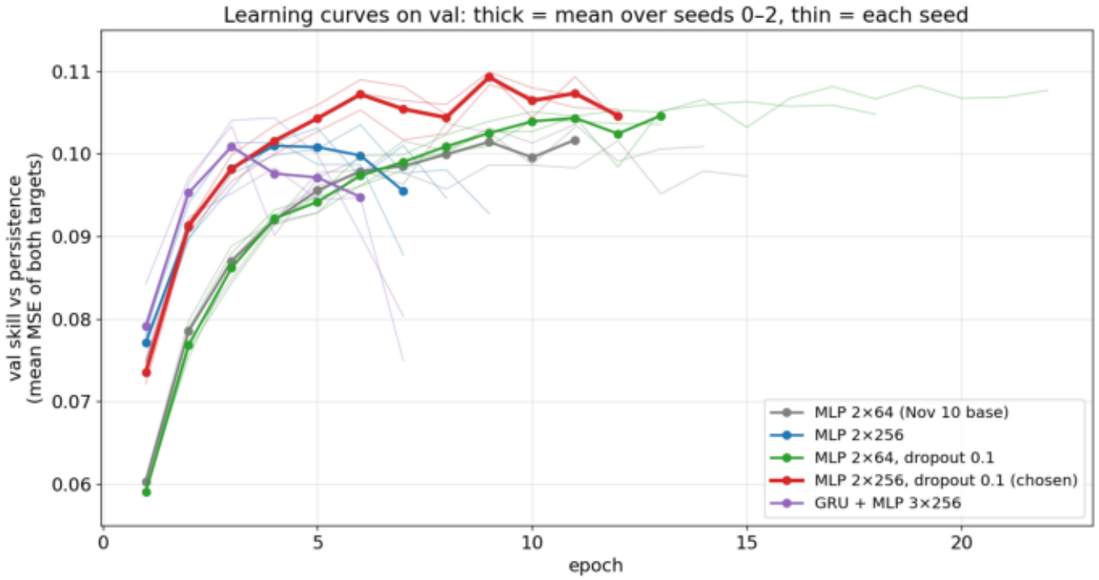
MLP 2×256 + dropout 0.1: val skill 0.109 ± 0.001 (classic), 0.109 ± 0.001 (e-bike)

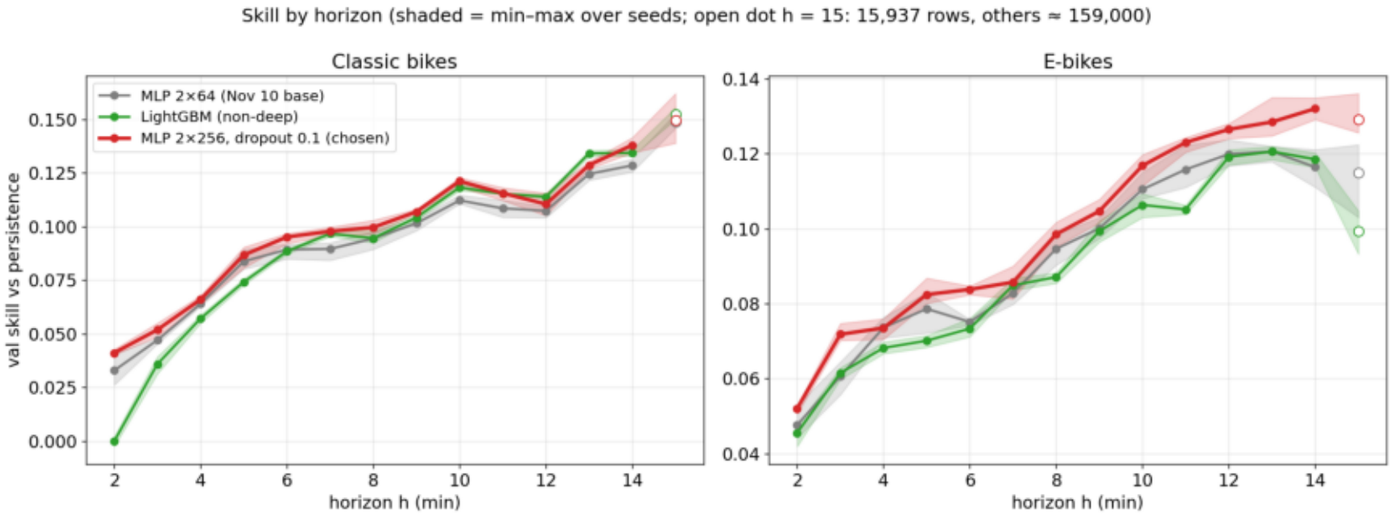


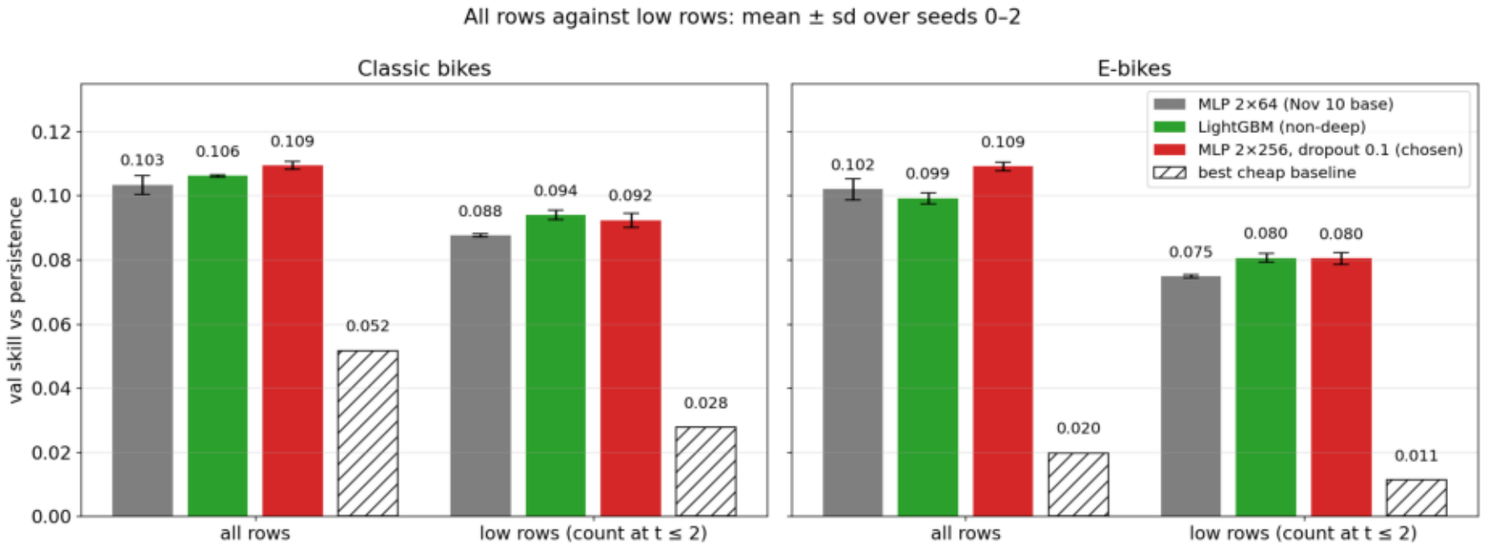
- Gain over the Nov 10 base: +0.006 / +0.007.
  - = 2.3 / 2.6 standard errors, counting seed noise and which val stations we got (bootstrap).
  - Positive on 5/5 val days (classic), 4/5 (e-bike).
- The gain is from dropout: without it, bigger nets overfit after 4–6 epochs.
- Width helps e-bikes only; on classic, 2×64 + dropout is as good.
- Dropout 0.3 scores best but is within noise on 3 of 4 columns and needs 2× the epochs; 0.5 drops.
- LightGBM is level on the low rows (-0.002 / -0.000).

Learning curves, horizon, low rows (val)

Dropout keeps improving after the plain nets overfit; the gain is largest for e-bikes at long h

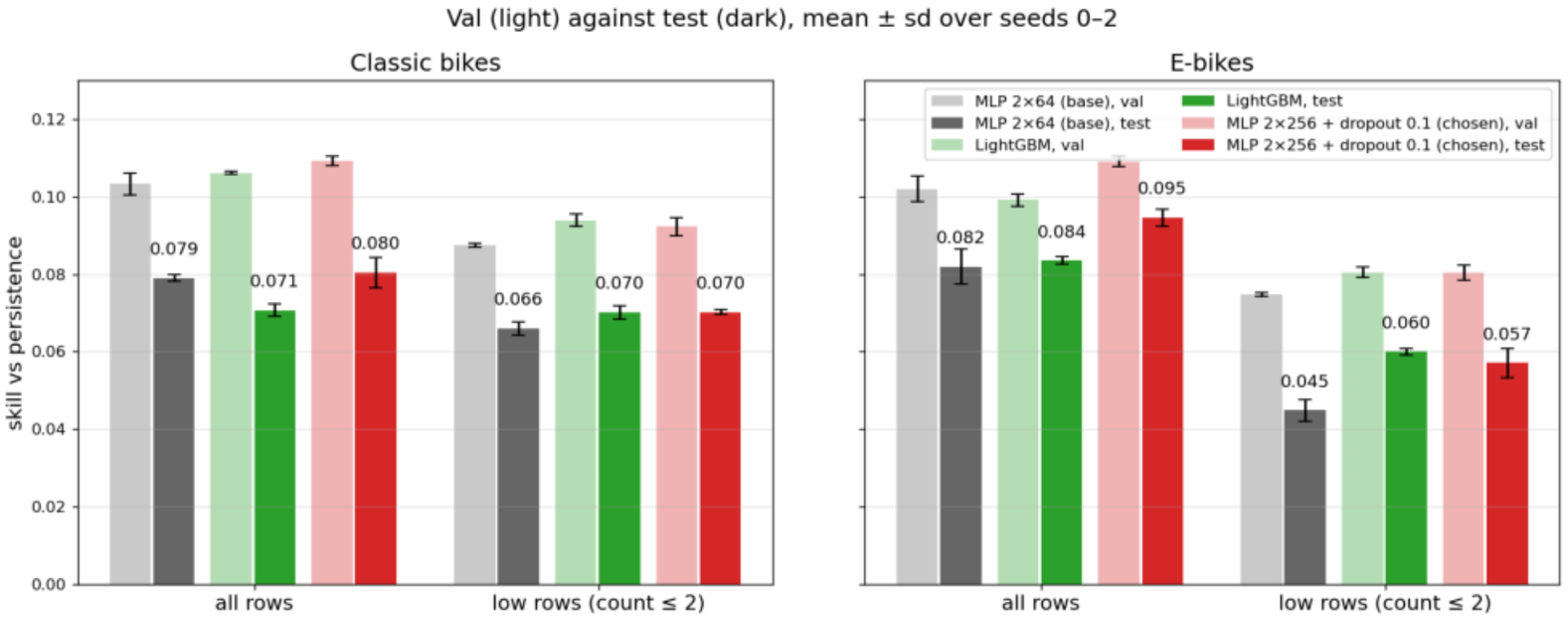






Test result (read once, protocol fixed before)

Chosen: 0.080 ± 0.004 (classic), 0.095 ± 0.002 (e-bike); test Oct 24–28, 1,747,660 rows

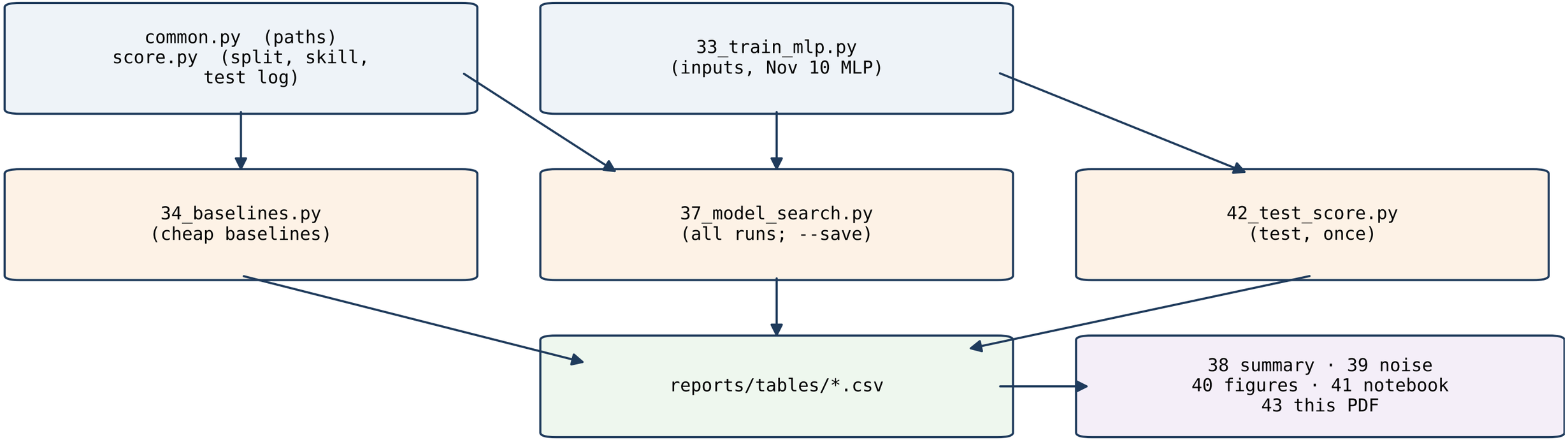


- E-bikes: the gain holds, +0.013 over the base and +0.011 over LightGBM.
- Classic: chosen 0.080 vs base 0.079 — the val gain (+0.006) does not carry over. Reviewers had flagged the classic gain as weak.
- Skill is lower than on val (0.109 / 0.109): test starts with a weekend, val has none. Chosen classic: 0.061, 0.034 on Sat / Sun, 0.114–0.117 on Mon / Tue, 0.005 on Wed (76,559 rows, a partial day).
- Low rows: chosen 0.070 ± 0.001 / 0.057 ± 0.004; LightGBM 0.070 ± 0.002 / 0.060 ± 0.001.
- Unseen stations: 2,324 test rows (0.13%) at 2 stations; without them the chosen skill is 0.081 / 0.095 (no change).
- Nothing was tuned after the test score.

Source: test_team_summary.csv, test_team_by_day.csv, test_scores.csv (42_test_score.py)

Code: verify it, change it

All in src/ of this repo · <GitHub link: to be added (repo not decided yet)>



- Reproduce (from src/, Python env from requirements.txt): python 37_model_search.py --step 1 (or --run NAME --budget full --seed 0 1 2); python 38_search_report.py --finalists …; python 39_val_noise.py; python 40_meeting_figures.py; python 42_test_score.py --dry (val checks only).
- Trace: notebooks/model_search_trace.ipynb (+ .html) prints every number of MODEL_SEARCH.md from its CSV with the script and commit; it checks the data SHA-1 and that the search never reads the test split.
- To change the model: add a config to GRIDS in 37_model_search.py; every run is logged to model_search_runs.csv. Data path: score.TEAM.

Decisions for the team, open questions

Recommendation first

- 1. Model for the progress report (Fri Nov 13, 11 pm): MLP 2×256 + dropout 0.1, with the base, LightGBM and the cheap baselines as references, val and test. Alternative: keep the Nov 10 base (test: same on classic, −0.013 on e-bikes).
- 2. Dropout rate for the final report: keep 0.1; settle 0.1 vs 0.3 on data not used for selection (e.g. a weekend check, Oct 17–18). The test split is used up for this choice.
- 3. Test protocol: done (3 seeds, mean ± sd, read once). Any later model gets its own single test score.
- 4. Report LightGBM beside the baselines: it is level on the low rows, so the deep model's lead is on e-bikes and on rows with 3+ bikes.

Open questions

- Larry (data): is the weather-fixed build (SHA-1 263bb791…) the team release? Does anyone still train on the older release with the weather leak?
- David (models): which models and numbers do you have? Can they be scored with score.py on the same split, for one table in the progress report?
- All: val has no weekend while test starts on one; add weekend days to val for the final? Unknown stations are rare on test (0.13 % of rows): leave the untrained row as is?